

Designing and Developing User Interfaces with AI: Advancing Tools, Workflows, and Practices

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Abstract

Designing and developing user-friendly interfaces has long been a cornerstone of HCI research. However, we are now at a turning point for how user interfaces can be designed and evaluated with new AI-based models and tools. The latest AI models have shown capabilities to model user behaviors, automate end-user tasks, and even generate user interfaces. We are at a pivotal moment to reflect on current UI design and development practices and discuss the opportunities, challenges, and risks brought by AI. Both incremental improvements and transformative opportunities exist for designers, developers, and the hand-off process in between. In this proposed workshop, we encourage participants to envision AI-enabled UI prototyping tools, workflows, and practices. By bringing together academic researchers and industry practitioners, we aim to identify opportunities to enhance prototyping tools and reshape UI creation workflows for the future as well as discuss potential negative consequences of these tools.

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1 Overview

AI models that can generate user interfaces (UIs) led us to imagine how AI might change the UI prototyping tools and practices [15, 16, 23, 26, 31, 54]. Their potential impact on prototyping tools range from incremental task acceleration to transformative workflow paradigm shifts [21, 40, 43]. For the past several years, research prototypes [5, 10, 11, 21] and commercial products (e.g. Figma AI, V0 by Vercel, Magic Patterns¹) have shown great promise in changing UI design and development practices. While early adopters have begun to experiment with these innovative AI tools, widespread adoption is to come. This transitional period offers a valuable opportunity for Human-Computer Interaction (HCI) researchers to reflect on existing changes enabled by AI and envision AI's ideal role in UI prototyping tools and workflows of the future².

Direct integration of AI into UI design and development tools is a result of years of HCI and AI research. Early research in data-driven approaches to understand UI semantics has built the foundation for UI generation with AI [38]. UI datasets such as Rico [6] and WebUI [60] have enabled training deep learning models on large-scale, diverse UI data. Various UI representation learning techniques have been proposed to utilize the multimodal nature of user interfaces [1, 28, 34, 57]. Recent large general-purpose, pre-trained multimodal AI models have enabled more sophisticated understanding and generation of UIs [22, 42, 56, 62]. For UI development, AI models' capabilities of generating UI code have also significantly improved over the years [2, 59, 61]. As AI lowers the barrier to writing UI code, it contains immense potential to transform both developer workflows and the designer-developer handoff process.

¹Figma AI: figma.com/ai, V0 by Vercel: v0.dev, Magic Patterns: magicpatterns.com

²In this proposal, we use the term "prototyping" to cover both UI design and development.

In addition to direct generation, AI has shown promise in evaluating and critiquing UI design. Research has indicated AI's technical feasibility to provide good-quality UI design critique through finetuning general purpose AI models [8, 9, 58]. Some work also explored using AI to annotate rationale for existing design examples [41]. Another important line of research in UI evaluation focuses on user behaviors: AI models that predict users' scanpaths using eyetracking data [17, 18] and user behavior modeling [53] showed alternative methods to understand end users. These new evaluation methods opened up new, low-cost ways to test UI prototypes, yet raises concerns around the truthfulness and bias in synthetic evaluation data.

Integrating AI for end user task automation also poses new requirements and challenges for UI design and development. Past HCI research has explored many ways to accelerate end users' task execution on UIs with AI [32, 33, 35, 36, 50, 52, 56]. These technical breakthroughs have been gradually integrated into real world products, shifting the balance between the two major interaction paradigms: direct manipulation and interface agent automation [51]. More tasks can now be executed by AI agents fairly reliably, yet the impact of AI agents on user agency and autonomy remains under-explored. Nevertheless, novel user experiences that potentially involve more interface agents create new challenges for UI design and development: how might we comprehensively design for unpredictable AI task automation scenarios? What are the UI safety concerns when agents act on users' behalves? How might we maximize user agency yet utilize AI agent efficiency gains? These are prime research questions to explore in our workshop.

The introduction of AI into end-user interactions and design and development workflows has brought about numerous new challenges; however, prototyping practices have largely been passively adapting to these changes [24, 29, 43]. This is a pivotal moment for the HCI community to examine evolving industry practices, draw insights from past experiences, and reimagine future UI design and development tools that effectively leverage AI.

UI prototyping tools are at an inflection point. Building upon previous workshops on design tooling [4] and AI approaches in UI [16, 20, 21] at CHI, this workshop aims to advance the frontier of AI-driven design and development. We invite academic and industry experts to explore future directions in technical algorithms, models, design and development workflows, and human-AI interaction paradigms. Discussion will emphasize preserving designers' and developers' **agency and creative control**, as well as end users' autonomy and safety with AI-created design and AI task automation. We advocate for enriched dialogues encompassing both underlying technology for UI prototyping supports and prototyping workflows that effectively leverage AI. The main topics of our workshop include:

- **AI Integration in UI/UX Design and Development Tools:** This topic explores direct integration of AI technologies into UI/UX design and development tools, examining the datasets and models that power these AI systems and how they can enhance design workflows by automating repetitive tasks, generating design suggestions, and optimizing layouts based on data-driven insights. The goal is to understand the implications of these integrations and identify best practices for

leveraging AI without compromising the creative input of designers and developers.

- **Agency and Creative Control in Evolving Workflows with AI:** AI integration in prototyping workflows offers opportunities to enhance efficiency and creativity, but it also poses challenges in maintaining the agency of designers and developers. This topic examines strategies for integrating AI that enhance, rather than hinder, creative control and decision-making. Emphasis will be placed on techniques that allow designers to interact with AI suggestions dynamically, ensuring that the human touch remains central to the design process, and AI acts as a supportive partner rather than a controlling force.
- **Safety Considerations in UI Automation:** As AI-driven automation becomes more prevalent in UI design, new interaction scenarios emerge that require careful consideration of user safety and privacy. This topic addresses the risks associated with AI-enabled automation, such as unintended user actions or security vulnerabilities. Discussions will focus on developing guidelines and best practices for implementing AI in ways that prioritize user safety, including transparency, accountability, and fail-safe mechanisms to protect end-users.
- **AI-Driven UI Evaluation Techniques:** Utilizing AI for UI evaluation involves employing computational models to simulate and analyze user behavior and interaction patterns. This topic covers methods such as predictive modeling of user actions, sentiment analysis from user interactions, and leveraging AI to perform usability testing at scale. The focus is on how these techniques can provide actionable insights into design performance, enabling more responsive, adaptive, and user-centered interface design.
- **Enhancing Designer-Developer Handoff with AI:** The designer-developer handoff, where designers transfer design assets to developers for implementation, is a critical phase in the product development cycle that is often fraught with challenges. This topic investigates how AI tools can facilitate a smoother transition of design assets and specifications, automatically translating design intent into code-ready components. By improving communication and reducing manual translation efforts, AI can help bridge the gap between design and development, fostering more collaborative and efficient workflows.

2 Background

2.1 UI Design and Development Tools, Practices, and Workflows

Since the emergence of personal computing platforms, HCI researchers and practitioners have been pushing forward our understanding of best prototyping practices (for both design and development) and innovation in tooling for prototyping. Early research explored critical aspects of user mental processes when interacting with technology, such as mental model, affordance, the gulfs of execution and evaluation, etc. [45, 48]. These insights have guided UI design and development practices ever since. Early digital design tools also emerged as HCI research prototypes [25], with later

research exploring model-based UI development methods to aid designers and developers in creating UIs [13, 46]. Researchers have also emphasized the value of sketching low-fidelity prototypes in early stages, as well as parallel prototyping to avoid design fixation [3, 7]. This research has influenced tools, practices, and workflows in the UI/UX industry for decades and significantly shaped the digital ecosystem people use and create nowadays. Our workshop carries this legacy forward, aiming to further explore the evolving landscape of UI prototyping by integrating cutting-edge AI technologies.

2.2 AI Advancements to Understand, Generate, and Evaluate UIs

Data-drive methods in machine learning have been applied to the domain of UI design and development. Large-scale datasets such as Rico [6], WebUI [60], and their extensions [27, 44, 57] have enabled deep learning explorations on a variety of UI-related tasks, including design generation [14, 30, 42], code generation [40, 59], UI modeling [1, 34, 62], and user behavior prediction [17, 18, 49]. AI models' impressive capabilities on these tasks opened up new opportunities to facilitate UI prototyping. In addition, the latest large pre-trained AI models pushed model capabilities further, including for UI-related tasks. Using certain prompting techniques, general-purpose large multimodal models have shown to perform well on tasks that require basic understanding of UI structure and content [56]. Fine-tuning pre-trained AI models can further improve their performance on tasks such as UI evaluation [8, 39, 58] and UI code generation [59]. Designing multi-agent workflows with pre-trained multimodal AI can also support sophisticated tasks such as UI accessibility [55] and UI design generation [12]. In this workshop, we encourage participants to discuss desired AI model capabilities, the required data and training techniques, and their applications in realistic design and development workflows.

2.3 Adopting AI Into AI Design and Development

While research systems and commercial products are gradually integrating AI into UI design and development workflows, the ideal role of AI in these workflows remain an open-ended research question. Previous survey has shown that UX professionals would like to see AI as a *creative partner* for inspirations and as an *assistant* to automate mundane tasks [24]. Research prototypes and commercial tools features often fall into these two main mechanisms of support. For example, researchers have created gallery systems to help designers find better inspirations using AI [10, 15]. Many generative UI mechanism have been proposed to automate mundane steps in UI design and development, while also kickstarting creative explorations [12, 40]. As AI lowers the barrier for both designing and developing good-quality user interfaces, a great potential touchpoint for AI integration is the designer-developer collaboration and handoff process [37]. However, how to effectively preserve user agency and creative control in AI-enabled prototyping tools remains an open research challenge and is currently left to specific design decisions [43]. Meanwhile, common concerns around AI, such as bias, trust, and copyright issues, still significantly impact its adoption in the UI/UX domain [24]. In our workshop, we seek to identify

ideal roles of AI in different stages of UI design and development and address the concerns alongside of it. We encourage participants to reflect on the bigger picture of design & development practices, imagining foundational paradigm shifts that can be brought on with new AI capabilities in both *inspiration* and *automation*.

3 The Goal of the Workshop

The primary objective of this workshop is to inspire the community to explore research opportunities in AI techniques and tools and their impact on design workflows and practices. Our aim is to have impact in academic, practical, and social contexts in designing and implementing high-quality, user friendly interfaces and experiences. We encourage individuals from various backgrounds, including the CHI community, adjacent academic fields, and industry practitioners to participate and think about future opportunities and challenges.

Through the discussions held during the workshop, our intention is to reflect on different ways AI will influence user interactions with interfaces and their implications on design and development requirements, tools, and practices. We draw attention to research on user behaviors, UI/UX design practices, and the latest AI capabilities to spark conversations between attendees and inspire participatory design of the preferred future state. We intend for this workshop to act as a platform that advances both new research directions and direct product impact.

4 Organizers

The organizing team has successfully organized three workshops at CHI on different topics of computational user interfaces [19–21]. The organizing team brings a wealth of successful experience in hosting CHI workshops on related topics. Before shifting to this proposal's specific focus on AI-driven design tools and workflows, members of the team successfully organized workshops on computational methods for understanding, generating, and evaluating user interfaces. Leveraging this expertise, the organizers are committed to delivering a high-quality and inspiring workshop experience, ensuring smooth planning and execution tailored to exploring future design tooling and workflows with AI.

The organizing team includes both academic and industry researchers, ensuring a balanced perspective that bridges cutting-edge research and practical applications in AI-supported design tooling. This blend of expertise allows the team to create a workshop environment that is relevant and engaging for participants from diverse backgrounds. Their collective experience will facilitate insightful discussions and foster collaborations that drive forward the research and practical application of AI-enhanced design tools.

Yuwen Lu is a Ph.D. student in the Department of Computer Science and Engineering at the University of Notre Dame, working on using data-driven approaches for understanding and generating user interfaces to support UX research and design work. Before joining Notre Dame, Yuwen received a Master's degree in Human-Computer Interaction from Carnegie Mellon University.

Yue Jiang is a Ph.D. student in Intelligent Systems supervised by Prof. Antti Oulasvirta at Aalto University and the Finnish Center for AI in Finland. Her research focuses on computational user interface

understanding, with specific interests in generating adaptive UIs for different users and contexts, AI-assisted design, and modeling human behavior.

Tiffany Kneareem is a User Experience Researcher on the Material Design team at Google. Her research focus is on product designer-developer collaboration, creativity support tooling and opportunities for AI in the user interface (UI) design space. She also conducts research on human-AI alignment from a human-centered perspective which places value on human agency to foster effective human-AI collaboration. She holds a PhD in Information Sciences and Technologies from Pennsylvania State University, advised by Dr. John M. Carroll.

Clara Kliman-Silver is a Staff UX Researcher at Google who studies design teams, design systems, UX tools, and designer-developer collaboration. She specializes in participatory design and generative methods to investigate workflows, understand designer-developer experiences, and imagine ways to create UIs. In previous roles, she has conducted research on developer tools, artificial intelligence, and healthcare. Clara holds a Bachelors of Science in Cognitive Science from Brown University.

Christof Lutteroth is a Reader in the Department of Computer Science at the University of Bath. His main research interests are in HCI, with a focus on immersive technology, interaction methods, and user interface design. In particular, he has a long-standing interest in methods for user interface layout. He is the director of the REal and Virtual Environments Augmentation Labs (REVEAL), the research center for immersive technology at the University of Bath.

Jeffery Nichols is a Research Scientist in the AI/ML group at Apple working on intelligent user interfaces. Previously he was a Staff Research Scientist at Google, working on the open-source Fuchsia operating system. His most important academic contribution recently was the creation of the RICO dataset [6]. He also worked on the PUC project [47], whose primary focus was creating a specification language that can define any device and an automatic user interface generator that can create control panels from this specification language.

Wolfgang Stuerzlinger is a Professor at the School of Interactive Arts + Technology at Simon Fraser University. His work aims to gain a deeper understanding of and to find innovative solutions for real-world problems. Current research projects include better 3D interaction techniques for Virtual and Augmented Reality applications, new human-in-the-loop systems for big data analysis, the characterization of the effects of technology limitations on human performance, investigations of human behaviors with occasionally failing technologies, user interfaces for versions, scenarios, and alternatives, and new Virtual/Augmented Reality hardware and software.

5 Pre-Workshop Plans

Before the workshop, we will distribute a call for participation across a variety of HCI-related emailing lists and social media, like Twitter and LinkedIn. The call will invite researchers and practitioners to contribute by submitting position papers. We will also

advertise the workshop at upcoming HCI conferences, among research groups, and through our professional networks. All participants are expected to submit a position paper. The submissions will be reviewed by the workshop organizers and committee members. The selection of participants will be based on the relevancy, innovation, and quality presented in their submissions according to workshop topics and criteria. To help candidates get familiar with the workshop's scope and goals, we have created a website <https://sites.google.com/view/computational-uichi25/home>, to provide information about the workshop.

6 ACCESSIBILITY

Authors whose position papers are accepted will be strongly encouraged to make their papers accessible. While they are preparing for the camera-ready version, our organizing team will help them with suggestions on how to make the documents accessible, like adding alt-texts for pictures and tables, and setting the order. To make sure the workshop is accessible to people with disabilities, we will consider adding subtitles, depending on what the participants need.

7 Workshop Structure

The workshop, scheduled for one day, will accommodate roughly 30 participants (including the organizers). The workshop will include two keynotes, presentations of paper and demo submissions to the workshop, and focused group discussions on a variety of related topics.

7.1 In-Person Format

The workshop is anticipated to be in-person. Standard equipment available at the conference center will suffice for technical requirements. The workshop website <https://sites.google.com/view/computational-uichi25/home>, will serve as a hub for information, hosting calls for papers, program details, organizers and speakers list, and pre-prints of accepted papers.

7.2 Workshop Schedule

Throughout the workshop, the attendees will engage with domain experts, and the organizers will guide discussions across various domains. The tentative agenda is shown in Table 1.

7.2.1 Keynotes. We will invite two keynote speakers who are experts currently working on AI-enabled design tools and workflows. Each will give a talk for 30 minutes, followed by an extensive Q&A and interactive discussion.

7.2.2 Paper & Demo Presentations. Accepted papers and demos will be categorized based on their themes for presentation. We will select the best two position papers in each category for a full presentation, each allotted a 10-minute slot. Some selected papers will have lightning talks with 1-minute slots. The duration and number of presentations will be adjusted as needed to accommodate the number of accepted submissions.

7.2.3 Breakout Group Discussions. After each presentation section, participants will be divided into smaller discussion groups. The

Time	Session
9:00 - 9:30	Introduction of workshop organizers, participants, topics, and goals
9:30 - 10:30	Keynote 1 by an invited speaker
10:30 - 11:00	Coffee break
11:00 - 12:00	Paper & Demo Presentation
12:00 - 12:30	Group discussion
12:30 - 13:30	Lunch
13:30 - 14:30	Keynote 2 by an invited speaker
14:30 - 15:30	Paper Presentation
15:30 - 16:00	Coffee break
16:00 - 17:00	Group discussion
17:00 - 17:30	Discussion group report back, wrap-up
17:30	Dinner (optional)

Table 1: Tentative agenda of the workshop

groups will be divided differently after each session to help participants get to communicate with more people. Participants can also suggest and create new groups based on their interests.

8 Post-Workshop Plan

After the CHI workshop, we plan to produce a report on the workshop outcome. The workshop papers and results will be available on the website before and after the workshop, providing opportunities for a larger audience to get updated on the events and results of our workshop. We may seek opportunities for an edited book or a special issue in a selected journal, *e.g.*, ToCHI, where the participants will be encouraged to publish their work.

A central goal of this workshop is community building for researchers and practitioners in this area. After the workshop, we plan to create a platform for community members to continue the discussion and share resources. Potential options may include a periodical email newsletter, a public GitHub repository, or a Slack/Discord channel. Participants and organizers will discuss the next steps at the workshop.

9 Call for Participation

“Designing User Interfaces and User Experiences with AI: Future Tools, Workflows, and Practices” is a workshop at CHI 2025. In this one-day workshop, our aim is to facilitate collaboration among researchers from various sub-disciplines of HCI, bridging the gaps between HCI and adjacent fields such as ML, CV, NLP, and Software Engineering. We welcome participants working on research and applications of AI in UI/UX design tooling and practices, from both industry and academia. Our primary goal is to encourage discussions regarding the future potential and requirements of computational approaches for user interfaces.

We invite researchers and practitioners to contribute by submitting a 4–6 page position paper in the double-column CHI Extended Abstract format (excluding references) to participate in the workshop. Alternatively, a video demo of less than 10 minutes can be submitted if the main idea can be more easily conveyed in this format. We will have a peer-review process, with each submission reviewed by at least two committee members or organizers. Selection for submission will be based on the criteria of quality and relevance.

Participants should follow the instruction on the website and submit the position papers via user.interface.workshop@gmail.com. Submissions can cover but are not limited to the following topics:

- **AI Integration in UI/UX Design and Development Tools:** We seek insights on integrating AI into design and development tools, focusing on datasets and models that enhance workflows by automating tasks, generating design suggestions, and optimizing layouts. Submissions should explore the benefits and best practices for AI integration while preserving designers’ and developers’ creative input.
- **Agency and Creative Control in AI-Enhanced Workflows:** We invite contributions on strategies that integrate AI into prototyping workflows without compromising designer and developer agency. Emphasis is on methods that keep human control central while enhancing creativity and efficiency.
- **Safety Considerations in UI Automation:** We welcome studies on safety and privacy challenges in AI-driven UI automation, focusing on risks like unintended actions and security vulnerabilities. Submissions should propose guidelines for safe AI implementation in design.
- **AI-Driven UI Evaluation Techniques:** Submissions are encouraged on AI methods for evaluating UIs, including predictive modeling, user interaction analysis, and large-scale usability testing. We seek approaches that offer actionable insights to improve design performance.
- **Enhancing Designer-Developer Handoff with AI:** We call for insights into AI tools that streamline the designer-developer handoff, reducing manual effort and improving collaboration. Submissions should address how AI can facilitate smoother transitions from design to development.

We will recruit researchers and practitioners in this field as program committee members to review submissions together. We will select submissions based on quality, novelty, and topic fit while aiming for a balance of different perspectives. Accepted papers will optionally be available on the workshop website (with the author’s consent). At least one author of each accepted position paper must register and attend the workshop and register for at least one day of the conference. The authors of each accepted position paper

will have about 8 minutes to present their work followed by an additional 2-minute Q&A.

9.1 Estimated Key Dates

- Call for participation released: December 15, 2024
- Position paper submission deadline: February 23, 2025
- Notification of acceptance: March 15, 2025
- Workshop date: April 26th, 2025

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